

REVISION 01 - STATEMENT OF WORK CONSTRUCTION

PROJECT: Installation of Oil Water Separator System

LOCATION: Hush House No. 2, Mountain Home AFB

NEXGEN WT: 14118044

~~DATE: July 28, 2025~~

DATE: May 5th, 2026

SCOPE OF WORK

- **Revised requirements for alarms to match ufgs**
- **Revised vent requirements**
- **Removed references to ufc 3-240-04A**
- **Removed references to IDEQ and NPDES discharge permit**
- **Added example products**
- **Frost depth clarification**

General:

The contractor shall furnish all personnel, equipment, supervision, and materials necessary to design, fabricate, install, and commission a complete and functional Oil Water Separator (OWS) system that will be fully integrated into the existing facility infrastructure. This system must handle wastewater from aircraft maintenance operations, ensuring compliance with EPA, DoD, and NPDES regulations. Updating this facility to meet these standards will allow a more efficient and continuous way to test jets and their engines by reducing maintenance intervals between uses and reducing possible environmental hazards.

EXECUTION PLAN

Assess suitable area, excavate and install into existing facility infrastructure a 300 GPM OWS, that meets or exceeds any applicable regulations.

Phase 1:

Pre-Construction Planning & Submittals

1. Material and Design Submittals

- a. Submit design drawings, product data, and specifications for but not limited to:
 - i. OWS system components
 1. Capable of a flow rate of 300 GPM (Gallons per minute)
 2. 99% Removal of free and dispersed hydrocarbons larger than 20 microns

3. Designed to allow gravitational and coalescing plate separation for optimal hydrocarbon phase separation.
 - ~~4. Must meet or exceed 40 CFR Part 112 (Oil Pollution Prevention)~~
 - ~~a. Must meet or exceed 40 CFR 122 (NPDES Permit Regulations)~~
 5. Example products that meet or exceed government expectations:
 - a. JB-25-64-300
 - b. 03000HGDWHTCHDS
 - c. HQB3064
- ii. Sensors and automation systems
1. Integrated sensors for real-time detection of oil accumulation beyond 15% tank volume threshold.
 2. Provide a visual alarm that illuminates in the event of a detected alarm condition. Include either individual lights for each alarm condition or include a single light and a liquid crystal display (LCD) panel that displaces information regarding each alarm condition. The visual alarm must be located in an area where it can be seen near the oil water separator when illuminated in the event of an alarm condition. ~~Smart alert for maintenance personnel when hydrocarbon level exceed environmental discharge limits.~~
 3. Furnish system that continuously and automatically monitors the interstitial space of an underground tank for breaches in the integrity of the inner and/or outer tank shells. Monitor the interstitial space by using either an electronic capacitance type liquid sensor or a positive pressure system.
 4. Scheduled solids extractions and dewatering system to prevent sludge overloading.
 - ~~5. Vapor phase carbon adsorption units mitigate volatile organic compound (VOC) emissions.~~
 6. Vent must conform with NFPA 30 and UL 142 and sized by tank manufacturer. ~~Explosion proof venting system for flammable vapor control.~~
- iii. Coatings and structural materials
1. OWS tank will be made from corrosion-resistant steel, stainless steel (316L), or fiber reinforced polymer (FRP) tank construction to withstand exposure to JP-8, MIL-PRF-7808, MIL-PRF-83282, and MIL-PRF-23699 aviation-grade fuel and lubricants.
 2. OWS interior will be epoxy-based, high-resistance coating for hydrocarbon and chemical exposure mitigation.
- iv. Control system schematics
- ~~1. Will design, produce and implement working schematic of operating control system for OWS.~~

2. Wiring schematics for all parts of the system showing each operating device and listing their normal ranges of operating values (including pressures, temperatures, voltages, currents, speeds, etc.).
 3. Single line diagrams of the entire system, including any required alterations to the existing piping and tank(s).
- b. All products must meet FAR Buy American compliance.
- c. All submittals are due within 20 calendar days of NTP.

2. Work Plan

- a. Timeline with major milestones (mobilization, construction, commissioning)
- b. Roles and responsibilities of project administration
 - i. During performance under contract, the contractor shall participate in administrative tasks inherent to the administration, surveillance, and oversight
- c. Role and Resume of Contract Employees
 - i. All contract personnel must be able to read, write, speak, and understand English fluently. Contract personnel shall present a neat appearance and be easily identified as a contract employee. The contractor shall obtain base identification, and vehicle passes for all employees who make frequent visits to or perform work on Mountain Home AFB in accordance with the current installation access/entry requirements. Contractor personnel are required to wear or prominently display installation identification while visiting or performing work on the installation. The contractor shall not employ persons for work on this contract, identified to the contractor by the CO as a potential threat to the health, safety, security, general well-being, or operational mission of the installation and its population. Each employee will be easily identified by wearing distinctive clothing bearing the name of the company or by wearing appropriate badges that contain the company name and employee name(s). The contractor shall not employ any person who is an employee of the Government, if employing that person would create a conflict of interest. The contractor is cautioned that off-duty active military personnel hired under this contract may be subject to permanent change of station, change in duty hours, or deployment. Military Reservists and National Guard members may be subject to recall to active duty. The abrupt absence of these personnel could adversely affect the service provider's ability to perform; however, their absence at any time shall not constitute an excuse for non-performance under this contract.
- d. Safety protocols and training certificates (OSHA 30, HAZMAT, etc.)

3. Disinfection and Commissioning Plan

- a. Methodology for system flush-out
- b. Hydrostatic and operational testing strategy
 - i. Hydrostatic Testing: Verify tank integrity with 72-hour static water load test.
 - ii. Flow Capacity Testing: Confirm compliance with designed flow rates.
 - iii. Oil Separation Efficiency Testing: Conduct controlled hydrocarbon injection test and measure separation performance against EPA Method 1664 (IF APPLICABLE)

Phase 2:

Site Preparation

1. Geotechnical Survey

- a. Verify subgrade strength and drainage profile
- b. Determine suitability for OWS weight and excavation impact
 - i. Conduct geotechnical soil analysis to determine subsurface stability and load-bearing capacity.

2. Excavation & Ground Stabilization

- a. Ensure maximum water level is below required frost depth per UFC ~~3-240-01~~ ~~3-240-04A~~
 - i. MHAFB frost depth is 32” **government will accept shallower frost depth to accommodate existing inlet outlet elevations and existing ground surface elevations as necessary**
- b. Ensuring a minimum of 48-inches of clearance below the frost line. Install compacted crushed stone base or reinforced slab based on Geotech results

Phase 3:

System Installation

1. Tank and Mechanical Components

- a. Install corrosion-resistant tank system (316L SS or FRP)
- b. Internal epoxy coatings per hydrocarbon exposure spec
- c. Integrate internal separation and retention system

2. Hydraulic & Piping Network

- a. Schedule 80 PVC/stainless piping with 0.5–1% gravity grade
 - i. Prevent degradation due to petroleum-based products and wastewater, to include hydraulic fluids, jet fuel (JP-8, JP-10, etc.), grease, oils, soaps, and detergents.
- b. Install pre-filtration debris screen and auto shut-off valves
 - i. Screen mesh to be <100 microns in filtration, to capture foreign object debris (FOD), jet blast residues, and metallic particulates.

- ii. Automatic inlet flow restrictors and emergency shutoff valves to regulate surge flow conditions during fuel spill events. Install double-wall piping in critical zones to mitigate potential leaks.
- 3. Electrical & Automation Controls
 - a. Class I, Div 2 electrical enclosures
 - i. Provide explosion-proof electrical enclosure in hazardous locations.
 - b. Remote monitoring and shutdown capability
 - i. Implement redundant fail-safe mechanism throughout the system, with the ability to pause, shut down and de-energize the entire system as needed.
 - c. Level sensors, VOC monitoring, and maintenance alerts.
 - i. Automatic Level & Hydrocarbon Detection Sensors:
 - 1. Integrated sensors for real-time detection of oil accumulation beyond 15% tank volume threshold.
 - ~~2. Smart alerts for maintenance personnel when hydrocarbon levels exceed environmental discharge limits.~~
 - ii. Automatic Sludge Removal System:
 - 1. Scheduled solids extraction & dewatering system to prevent sludge overloading.
 - ~~iii. Secondary Vapor Control Measures:~~
 - ~~1. Vapor phase carbon adsorption units to mitigate volatile organic compound (VOC) emissions.~~
 - ~~2. Explosion-proof venting system for flammable vapor control.~~

Phase 4:

Commissioning & Testing

- 1. System Testing
 - a. 72-hour hydrostatic leak testing
 - i. Verify tank integrity with a static water load test.
 - b. Flow capacity validation (300 GPM)
 - i. Confirm designed flow capacity of 300 GPM
 - c. Oil separation test per Method 1664 (if applicable)
 - i. Conduct controlled hydrocarbon injection test and measure separation against established standards.
- ~~2. Environmental Compliance~~
 - ~~a. Conduct NPDES compliance validation for effluent discharge quality.~~
 - ~~b. Obtain final discharge permits and agency signoffs.~~
 - ~~i. Secure necessary State-level discharge permits if applicable.~~
 - ~~c. Submit compliance certificate to base CE and contracting office~~
- 3. Training & Documentation
 - a. On-site training for operator teams

- i. Provide comprehensive operator training on system operation, emergency shutdown, and routine maintenance.
- b. Submit “As-Built” drawings and O&M manuals
- c. Provide 1-year no-limit commercial warranty

TECHNICAL CRITERIA

1. All work shall conform to the latest editions of the following standards:
 - a. UFC 3-240-01 (Wastewater Collection and Treatment, with Change 4) ~~3-240-04A (Oil-Water Separator Design)~~
 - b. 40 CFR Part 112 and 122 (EPA Regulations)
 - ~~c. NPDES Permit Requirements~~
 - d. Federal Acquisition Regulation (FAR) Buy American Compliance
 - e. NFPA and NEC standards for hazardous environment installation
 - f. Idaho State Environmental Codes
 - g. 01 57 19 Temporary Environmental Controls_MHAFB Env Spec 20220614 (Attachment 1)

GENERAL PROVISIONS

1. Work Authorization: All work must be directed in writing by the Contracting Officer.
Hours of Operation: Monday to Friday, 0700–1630 MST. Federal holidays excluded.
2. Emergency Contact Numbers:
 - a. Fire Department: 911
 - b. CES PM: Brad Newcomb – 208-828-2792
 - c. WFSM R&O SME: Donoso, Samuel SSGT – 208-828-4015
 - d. WFSM Shop: 208-828-6538

ACCEPTANCE CRITERIA

1. The OWS system will be deemed successfully installed and operational when it meets the following criteria:
 - a. Achieves >99% hydrocarbon removal efficiency under load conditions.
 - b. Passes all hydrostatic, flow rate, and ~~EPA~~ discharge compliance tests
 - c. It is fully integrated into the facility without leaks or disruptions
 - d. Meets all safety, operational, and documentation requirements